

# Introduction to Internet of Things

Complete Lecture Notes — All 12 Weeks · 60 Lectures

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# WEEK 1 — IoT Foundations — Sensing, Actuation & Networking Basics

## Lecture 1 · Introduction to IoT — Part I

### What is the Internet of Things?

The Internet of Things (IoT) is an expansion of the current internet — which connects computers — to include physical objects ('things'): lights, fans, air-conditioners, refrigerators, toothbrushes, industrial machines, and everything in between. Each thing is fitted with **embedded electronics, a processor, and a communication module**, allowing it to act as a network node in a massive internetwork.

The fundamental vision is **anytime, anyplace, anything** connectivity. Estimates project **20 billion+ connected devices** in the near future, eventually reaching trillions.

### Two Approaches to Building IoT

|                                      |                                                                    |
|--------------------------------------|--------------------------------------------------------------------|
| <b>Expand the existing internet</b>  | Connect things to the current internet of computers.               |
| <b>Build a separate internetwork</b> | Create a brand-new network from scratch for physical objects only. |

### Enabling Technologies of IoT

- **RFID** — Radio Frequency Identification tags and readers.
- **Sensors and Actuators** — most essential components for building IoT.
- **Networking** — WiFi, Zigbee, Bluetooth, cellular, LoRa, 6LoWPAN.
- **Nanotechnology** — Internet of Nano Things (IoNT): nano-sized swallowable capsules forming nano-networks inside the body.
- **Supporting technologies** — Cloud computing, Big Data, Machine Learning, AI, Computer Vision.

### Key Characteristics of IoT

| Characteristic            | Description                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------|
| Efficiency                | Must serve application requirements effectively                                                   |
| Scalability               | Must work with billions/trillions of nodes without performance degradation                        |
| Addressing                | New naming/addressing schemes needed — IPv4 insufficient; IPv6 explored; new scheme likely needed |
| Low Power                 | Nodes are resource-constrained; use sleep/wake duty cycles                                        |
| Mobility                  | Nodes (smartwatches, vehicles) and sub-networks move; handled via opportunistic networks          |
| Intermittent Connectivity | Network partitions; addressed by opportunistic mobile networking                                  |

### Application Domains

| Domain                   | Market Share | Key Examples                                             |
|--------------------------|--------------|----------------------------------------------------------|
| Manufacturing & Business | ~40.2%       | Supply chain, robotic machinery, smart factories         |
| Healthcare               | High         | Remote monitoring, telemedicine, electronic records      |
| Retail                   | Significant  | Inventory tracking, smart purchasing, consumer analytics |

|          |         |                                                            |
|----------|---------|------------------------------------------------------------|
| Security | Growing | Biometrics, facial/fingerprint recognition, remote sensors |
|----------|---------|------------------------------------------------------------|

## Evolution of Connected Devices

ATMs (1970s) → Web (1990s) → Smart Meters (2000s) → Digital Locks → Smart Healthcare → Connected Vehicles → Smart Cities → **Smart Dust** (computers smaller than a grain of sand that can be spread anywhere to measure chemicals or diagnose problems in the human body).

## IoT vs M2M vs Web of Things

| Concept                  | Scope                                  | Key Focus                                                                       |
|--------------------------|----------------------------------------|---------------------------------------------------------------------------------|
| M2M (Machine-to-Machine) | Narrower — machine-to-machine only     | Communication between two devices; uses cloud/internet; a subset of IoT         |
| IoT                      | Broader — things, people, applications | Large-scale networking of physical objects; includes analytics, services, cloud |
| Web of Things (WoT)      | IoT + web layer                        | Uses HTML5, JavaScript, Ajax, PHP on top of IoT for web accessibility           |

## Lecture 2 · Introduction to IoT — Part II

### The Address Crunch Problem

IoT will have billions/trillions of devices. IPv4 provides only  $2^{32} \approx 4.3$  billion addresses — completely insufficient. IPv6 offers  $2^{128}$  addresses. Most current IoT still uses IPv4, with translation mechanisms bridging the gap until a new scheme is devised.

| Feature         | IPv4                         | IPv6                       |
|-----------------|------------------------------|----------------------------|
| Address length  | 32 bits                      | 128 bits                   |
| Address space   | ~4.3 billion                 | $\sim 3.4 \times 10^{38}$  |
| Notation        | Dotted decimal (192.168.1.1) | Hexadecimal (2001:0db8::1) |
| IoT suitability | Inadequate                   | More suitable              |

### IoT Network Components

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| <b>IoT LAN</b>     | Local short-range communication (building/campus-wide)                 |
| <b>IoT WAN</b>     | Connecting multiple LANs across geographical/organizational boundaries |
| <b>IoT Node</b>    | Individual device/thing within a LAN or WAN                            |
| <b>IoT Gateway</b> | Connects LAN to the internet; assigns locally unique addresses         |
| <b>IoT Proxy</b>   | Bridges internal (locally-unique addresses) to external internet       |

### Address Conservation & Mobility

- Nodes within a gateway domain have **locally unique addresses** — unique only within that LAN.
- The gateway has a unique **global prefix** assigned by its router.
- When a node moves to another network, the gateway prefix changes (Prefix 1 → Prefix 2) while the node's local address remains the same.
- **Remote Anchor Point**: A global-view entity that manages address changes during mobility.
- **Tunneling (IKEv2)**: Allows nodes to communicate through a remote anchor point.

- **Multi-homing:** Node connected to multiple networks simultaneously for improved reliability.

## Lecture 3 · Sensing

### What is Sensing?

A sensor detects changes in ambient conditions or the state of another device, then converts those physical phenomena into **electrical signals** for processing and transmission.

### Types of Sensors

#### By Output Type

| Type    | Description                   | Example                                           |
|---------|-------------------------------|---------------------------------------------------|
| Analog  | Continuous output             | Temperature sensor — continuously varying voltage |
| Digital | Discrete output (0/1, ON/OFF) | PIR sensor — HIGH when motion detected            |

#### By Data Type

| Type   | Measures                              | Example                                   |
|--------|---------------------------------------|-------------------------------------------|
| Scalar | Magnitude only — direction irrelevant | Temperature, pressure, color, humidity    |
| Vector | Magnitude AND direction/orientation   | Accelerometer (x,y,z axes), camera sensor |

### Sensors vs. Transducers

**Transducer** is a collective term that includes both sensors and actuators — it converts one form of energy into another (e.g., microphone: sound → electrical signal). A **sensor** specifically detects a physical property and is sensitive *only* to that property (e.g., a temperature sensor cannot detect smoke).

### Sensor Errors

| Error Type          | Description                                                                    |
|---------------------|--------------------------------------------------------------------------------|
| Sensitivity error   | Actual sensitivity differs from specified sensitivity under real conditions    |
| Offset / Bias error | Constant additive error at all times (e.g., always reads 2°C too high)         |
| Non-linearity       | Deviation from ideal linear transfer function over the full measurement range  |
| Drift               | Slow change in output independent of the measured property, over time          |
| Noise               | Random deviation due to external environmental factors                         |
| Hysteresis          | Present reading depends on past inputs (metal strip sensors, magnetic sensors) |
| Quantization error  | Approximation error in digital-output sensors                                  |
| Aliasing error      | Occurs when sampling frequency is insufficient (Shannon-Nyquist theorem)       |

### Common Sensors by Application

| Measurement          | Sensor Types                                |
|----------------------|---------------------------------------------|
| Light / Illumination | LDR (Light Dependent Resistor), Photodiodes |
| Temperature          | Thermistor, Thermocouple                    |
| Force / Pressure     | Strain gauge, Pressure switch               |

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| Position               | Potentiometer, Encoders, Opto-couplers                    |
| Speed                  | Reflective sensors, Doppler effect sensors                |
| Sound                  | Carbon microphone, Piezoelectric crystals                 |
| Chemical               | Liquid chemical sensors, Gas sensors                      |
| Distance               | Ultrasonic sensor (emits sound, measures round-trip time) |
| Motion / Obstacle      | PIR (Passive Infrared) sensor                             |
| Image / Visual         | Camera sensor (vector sensor)                             |
| Temperature + Humidity | DHT11, DHT22 (digital combo sensors)                      |

## Lecture 4 · Actuation

### What is Actuation?

Based on sensor readings, **actuators** perform physical actions on the environment or system. An actuator requires: (1) a **control signal**, and (2) a **source of energy**. Upon receiving a control signal, it converts that energy into mechanical motion.

### Types of Actuators

| Type               | Principle                                                                                                    | Examples                                                             |
|--------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Hydraulic          | Uses fluid pressure (oil/water). Near-incompressible liquids exert huge force.                               | Hydraulic cylinder, radial engine actuator                           |
| Pneumatic          | Compressed air/vacuum → linear or rotary motion. Very responsive to small pressure changes.                  | Pneumatic brakes (trucks), rack-and-pinion valve actuators           |
| Electrical         | Motor converts electrical energy to mechanical torque.                                                       | Solenoid valve (water flow control), electric bell, rotary actuators |
| Thermal / Magnetic | Shape Memory Alloys (SMAs) bend/deform when heated. Compact, lightweight, high power density.                | SMA piezo motor, coil gun (Hyperloop principle)                      |
| Mechanical         | Convert rotary motion to linear motion using gears, pinions, rails, pulleys.                                 | Rack and pinion, crankshaft                                          |
| Soft Actuators     | Polymer-based (SMPs). Respond to light, electrical, magnetic, heat, pH stimuli. For fragile object handling. | Fruit harvesting robots, biomedical internal organ handling          |

## Lecture 5 · Basics of IoT Networking — Part I

### Complete IoT System Architecture

A complete IoT system has five key stages:

- **Sensors / Actuators** — physical nodes with sensing/actuating devices.
- **Local Network** — data flows from sensor nodes through a LAN (ZigBee, WiFi, Bluetooth).
- **Internet** — data forwarded from local gateway to backend servers.
- **Backend Services** — analytics, ML algorithms, neural networks run on cloud servers.
- **Actuation Response** — decisions trigger physical actuators (pump, valve, light, etc.).

## IoT Service-Oriented Architecture (4 Layers)

|                        |                                                   |
|------------------------|---------------------------------------------------|
| <b>Sensing Layer</b>   | RFID, sensors, data acquisition devices           |
| <b>Network Layer</b>   | Sensor networks, databases, internet connectivity |
| <b>Service Layer</b>   | Service integration, business logic, repositories |
| <b>Interface Layer</b> | Application frontend, APIs, contract interfaces   |

*Security spans all four layers as a horizontal concern.*

## Key IoT Networking Challenges

- Scalability and energy efficiency in large networks.
- Bandwidth management across heterogeneous link types.
- Interfacing and interoperability between devices from different vendors.
- Interference management in densely deployed wireless IoT networks.
- Heterogeneity in devices, protocols, operating systems, and algorithms.
- Data storage, analytics, and complexity management (SDN can help here).

# WEEK 2 — IoT Protocols & Connectivity Technologies

## Lecture 6 · Basics of IoT Networking — Part II: MQTT & SMQTT

### MQTT — Message Queue Telemetry Transport

MQTT is an **ISO standard publish-subscribe messaging protocol** originally developed by IBM (1999) and standardised by OASIS in 2013. It runs over TCP/IP and is designed to be lightweight for resource-constrained devices.

### Three Core Components

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Publishers</b>  | Sensors/devices that generate and publish data to the broker                            |
| <b>Broker</b>      | Middleware that receives published data, classifies by topic, dispatches to subscribers |
| <b>Subscribers</b> | Applications/devices that subscribe to topics and receive matching data                 |

### MQTT Methods

| Method      | Function                                |
|-------------|-----------------------------------------|
| Connect     | Connect to MQTT server                  |
| Disconnect  | Disconnect from server                  |
| Subscribe   | Subscribe to a topic                    |
| Unsubscribe | Cancel subscription                     |
| Publish     | Publish data to broker for distribution |

### Topics and Wildcards

A topic is a hierarchical string separated by '/', e.g., **house/livingroom/temperature**.

- **Single-level wildcard (+):** house/+/temperature matches house/livingroom/temperature AND house/kitchen/temperature.
- **Multi-level wildcard (#):** house/# matches all topics beginning with 'house/'.

### Event-Driven Architecture

MQTT is **event-driven** (not request-response like HTTP). Data is pushed to clients on events. Producers and consumers are fully decoupled — they only need to know the topic, not each other. This makes MQTT highly scalable.

### Real-World Applications of MQTT

- Facebook Messenger
- Amazon AWS IoT Core
- Microsoft Azure IoT Hub
- EVERYTHING IoT Platform
- Adafruit IO

### SMQTT — Secure MQTT

SMQTT extends MQTT with broadcast encryption (analogous to HTTP → HTTPS). One message is encrypted and delivered to multiple nodes simultaneously.

| Stage      | Description                                                              |
|------------|--------------------------------------------------------------------------|
| Setup      | Subscribers/publishers register with broker; receive a master secret key |
| Encryption | Data encrypted by publisher using agreed algorithm                       |
| Publish    | Encrypted data published to broker and dispatched                        |
| Decryption | Subscriber decrypts using the master secret key                          |

## Lecture 7 · Basics of IoT Networking — Part III: CoAP & XMPP

### CoAP — Constrained Application Protocol

CoAP is a **lightweight equivalent of HTTP** designed for resource-constrained IoT devices (limited power, bandwidth, computation). It is a session/application layer protocol running over **UDP** (not TCP). Based on IETF RESTful Environment Working Group.

#### CoAP Sub-Layers

|                                   |                                                     |
|-----------------------------------|-----------------------------------------------------|
| <b>Messaging sub-layer</b>        | Handles reliability and duplicate message avoidance |
| <b>Request-Response sub-layer</b> | Handles actual GET/PUT/POST/DELETE communication    |

#### CoAP Message Types

| Type                  | Description                                                                 |
|-----------------------|-----------------------------------------------------------------------------|
| Confirmable (CON)     | Requires ACK from the server                                                |
| Non-confirmable (NON) | No acknowledgement required — fire and forget                               |
| Piggyback             | ACK and response data sent together in one message                          |
| Separate              | ACK sent immediately; actual response data sent later as a separate message |

### XMPP — Extensible Messaging and Presence Protocol

XMPP is an **open-standard, XML-based, message-oriented middleware** for real-time structured data exchange. It uses a decentralised client-server model — anyone can run an XMPP server. Works through firewalls and virtual machine environments where alternatives like ping protocol fail.

#### Core XMPP Technologies

| Technology            | Function                                             |
|-----------------------|------------------------------------------------------|
| Core                  | XML streaming — base communication                   |
| Jingle                | Multimedia signalling: voice, video, file transfer   |
| Multi-User Chat (MUC) | Multi-party communication                            |
| Pub-Sub               | Publish-subscribe alerts and notifications           |
| BOSH                  | HTTP binding for XMPP (works through HTTP firewalls) |

#### XMPP Weaknesses

- Does not support QoS.
- Text-based → higher network overhead.
- Binary data must be Base64-encoded before transmission.



## Lecture 8 · Basics of IoT Networking — Part IV: AMQP

### AMQP — Advanced Message Queuing Protocol

AMQP is an **ISO/IEC 19464 standard application-layer protocol** designed for inter-business and inter-system messaging. Unlike MQTT (which targets small IoT devices), AMQP targets enterprise environments. Data unit: **Frame**.

#### Architecture: Producers → Server (Broker) → Consumers

|                 |                                                                  |
|-----------------|------------------------------------------------------------------|
| <b>Exchange</b> | Receives messages and routes them to the appropriate queues      |
| <b>Queues</b>   | Separate queues per business process; consumers read from queues |
| <b>Bindings</b> | Rules determining routing — who receives which messages          |

#### Four Exchange Types

| Exchange | Routing Behaviour                              |
|----------|------------------------------------------------|
| Direct   | Message routed to a specific, named queue      |
| Fan-out  | Message broadcast to all bound queues          |
| Topic    | Message routed based on topic pattern matching |
| Header   | Message routed based on header attributes      |

#### Message Delivery Guarantees

| Guarantee     | Meaning                                                               |
|---------------|-----------------------------------------------------------------------|
| At most once  | Delivered once or never — possible loss, no duplicates                |
| At least once | Always delivered, but may arrive more than once (duplicates possible) |
| Exactly once  | Guaranteed delivery of exactly one copy — highest reliability         |

#### Nine AMQP Frame Types

Open, Close, Begin, End, Attach, Detach, Transfer, Flow, Disposition. These initiate, control, and tear down message transfer between two peers.

## Lecture 9 · Connectivity Technologies — Part I: 802.15.4 & ZigBee

### IEEE 802.15.4

IEEE 802.15.4 is the foundational standard for **Wireless Personal Area Networks (WPANs)**. Focus: low data rate, low power, monitoring and control. Covers **Physical (PHY) and MAC layers** only. Operates in the ISM band. Used as the basis for ZigBee, 6LoWPAN, WirelessHART, and others.

| Parameter          | Detail                                                           |
|--------------------|------------------------------------------------------------------|
| Modulation         | DSSS (Direct Sequence Spread Spectrum) — noise-tolerant          |
| Low speed variant  | BPSK modulation                                                  |
| High speed variant | O-QPSK modulation                                                |
| MAC protocol       | CSMA-CA (Carrier Sense Multiple Access with Collision Avoidance) |

|                  |                                                        |
|------------------|--------------------------------------------------------|
| Min. power level | −3 dBm (0.5 microwatts)                                |
| Duty cycle       | < 1% (very energy efficient)                           |
| Indoor range     | 10–40 m                                                |
| Outdoor range    | Up to 75 m typical, 1000 m theoretical (line-of-sight) |
| Topologies       | Star and Mesh                                          |

### Device Types

|                                      |                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------|
| <b>FFD (Full-Function Device)</b>    | PAN coordinator, router, or general device. Supports all functions including routing. |
| <b>RFD (Reduced-Function Device)</b> | Can only communicate with an FFD. Minimal CPU/RAM. Low power. Cannot route.           |

### Network Types

|                           |                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Beacon-Enabled</b>     | PAN coordinator broadcasts periodic beacon frames. Uses slotted CSMA-CA. Nodes synchronise to beacons. |
| <b>Non-Beacon-Enabled</b> | Uses un-slotted CSMA-CA. Beacons only for link discovery. Requires source and destination IDs.         |

### 802.15.4 Variants (Regional/Application)

| Variant | Use                                 |
|---------|-------------------------------------|
| a, b    | Base standard                       |
| c       | China                               |
| d       | Japan                               |
| e       | Industrial applications             |
| f       | Active RFID                         |
| g       | Smart utility networks / Smart Grid |

## ZigBee

ZigBee works **on top of IEEE 802.15.4** (layers 1 and 2), extending functionality to **Layer 3 and above** (up to application layer). It adds authentication, encryption, and mesh routing. Heavily used in Wireless Sensor Networks (WSNs).

### ZigBee Topologies

- **Star:** Coordinator at centre + end devices in direct single-hop communication.
- **Cluster Tree:** Routers form a tree structure with coordinator at root.
- **Mesh:** Routers form a mesh backbone; end devices attach to routers. Most reliable — self-configuring and self-healing.

### ZigBee Entities

| Entity                  | Role                                                                                                           |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| ZigBee Coordinator (ZC) | Root of the network tree; single coordinator per network; stores network info; trust centre and key repository |
| ZigBee Router (ZR)      | Runs applications and relays data between connected nodes                                                      |

|                         |                                                                                 |
|-------------------------|---------------------------------------------------------------------------------|
| ZigBee End Device (ZED) | Minimal functionality; communicates only with parent node; cannot relay packets |
|-------------------------|---------------------------------------------------------------------------------|

### ZigBee Routing: AODV Protocol

ZigBee uses **AODV (Ad hoc On-Demand Distance Vector)** routing at the network layer. A route request is broadcast; intermediate nodes relay it until the destination is found. The destination then sends back a unicast route reply with the optimal path. Mesh networks are self-configuring and self-healing — if a node fails, alternative paths are used.

## Lecture 10 · Connectivity Technologies — Part II: 6LoWPAN & RFID

### 6LoWPAN — IPv6 over Low-Power WPANs

6LoWPAN stands for **IPv6 Low-Power Wireless Personal Area Network**. It allows the smallest IoT devices to transmit wirelessly using IPv6 addressing, bridging 802.15.4 radios with the internet through **header compression and address translation**. Based on IETF RFC 5933 and RFC 4911.

### Addressing in 6LoWPAN

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
| 16-bit short address    | PAN-specific; assigned by PAN coordinator; for intra-PAN communication |
| 64-bit extended address | Globally unique; for internet-wide addressing                          |

### 6LoWPAN Header Types

| Header                 | Purpose                                                                            |
|------------------------|------------------------------------------------------------------------------------|
| Dispatch Header        | Identifies next header type (6-bit dispatch field + 2-bit type)                    |
| Mesh Addressing Header | Stores originator, final destination, and hops-left counter (decremented each hop) |
| Fragmentation Header   | For first fragment (no offset) and subsequent fragments (includes datagram offset) |

### Routing Protocols in 6LoWPAN

|               |                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LOADng</b> | Lightweight On-demand Ad hoc Distance-vector routing. Derived from AODV. PDUs: Route Request, Route Reply, Route Error. Only the destination responds to route requests — intermediate nodes are prohibited from responding.                                                                |
| <b>RPL</b>    | Routing Protocol for Low-power and Lossy Networks. Distance-vector routing. Maintains topology via low-rate beaconing. Two modes: Proactive (maintain topology) and Reactive (resolve inconsistencies). Objectives: minimise energy, minimise latency, satisfy bandwidth/power constraints. |

### RFID — Radio Frequency Identification

RFID encodes data in small **tags** that can be read wirelessly by **RFID readers**. Predecessor: AIDC (wired). RFID uses radio waves — fully wireless. Similar concept to barcodes/QR codes but wireless.

### Tag Types

|                              |                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Passive tags (common)</b> | No internal power. Use magnetic induction from reader's electromagnetic field to power themselves and transfer data. |
| <b>Active tags</b>           | Have their own power source. Longer range and more features.                                                         |

### RFID Applications

- Inventory management and asset tracking.
- Attendance tracking and access control to restricted areas.
- ID badging and smart cards in organisations.

- Supply chain management and pharmaceutical counterfeit prevention.

## WEEK 3 — Connectivity Technologies (III-V) & Sensor Networks (I-II)

### Lecture 11 · Connectivity Technologies — Part III: HART, WirelessHART & NFC

#### HART — Highway Addressable Remote Transducer

HART is derived from IEEE 802.15.4 and is designed primarily for **Industrial IoT (IIoT)**. Wired HART does **not** have a network layer. **Wireless HART** extends it with networking capabilities and operates in the 2.4 GHz ISM band using 15 channels.

#### Wireless HART Protocol Stack

| Layer       | Detail                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------|
| Physical    | Based on 802.15.4; 2.4 GHz ISM band; 15 channels                                                 |
| Data Link   | Super-frames (10 ms slots), TDMA, channel hopping after every message, channel blacklisting      |
| Network     | Mesh topology; updated network graph for routing; handles OSI network+transport+session combined |
| Transport   | Handles diverse traffic types; part of combined network layer above                              |
| Application | Command/response messages; identical for wired and wireless HART variants                        |

#### Key WirelessHART Features

- ◆ Channel hopping: frequency changes after every message → improved security and reliability.
- ◆ Channel blacklisting: identifies and removes persistently-interfered channels.
- ◆ Super-frames: 10 ms TDMA slots ensure collision-free, deterministic communication.
- ◆ Network Manager Agent: supervises every node; guides when and where to send packets.
- ◆ Backward compatible: legacy wired HART devices can coexist with Wireless HART nodes.

#### ZigBee vs. Wireless HART Comparison

| Feature                | ZigBee                     | Wireless HART                       |
|------------------------|----------------------------|-------------------------------------|
| Primary use            | Consumer IoT               | Industrial IoT                      |
| MAC protocol           | CSMA/CA (contention-based) | TDMA (time-slotted)                 |
| Channel hopping        | Only entire network hops   | Every node hops after every message |
| Network topology       | Tree                       | Mesh                                |
| Backward compatibility | Limited                    | Yes — supports legacy devices       |
| Security               | Standard                   | Code-based network security         |

#### NFC — Near Field Communication

NFC is very similar to RFID — both use **magnetic induction**. Designed for devices within very close proximity. NFC reader emits a small electric current creating a magnetic field that bridges the physical space; the NFC tag/card picks up this field and transfers its encoded information back.

| Parameter              | Value                           |
|------------------------|---------------------------------|
| Operating frequency    | 13.56 MHz                       |
| Data rates             | 106 Kbps, 212 Kbps, or 424 Kbps |
| Communication range    | < 20 cm                         |
| Storage (passive tags) | 96–512 bytes                    |

### NFC Modes of Operation

| Mode           | Description                                            | Example                                 |
|----------------|--------------------------------------------------------|-----------------------------------------|
| Peer-to-peer   | Two active NFC devices exchange data directly          | Two smartphones sharing contact or file |
| Read/Write     | Active device reads information from a passive NFC tag | Smartphone scanning NFC poster or label |
| Card Emulation | NFC device acts as a contactless smart card            | Contactless credit card payment         |

## Lecture 12 · Connectivity Technologies — Part IV: Bluetooth

### Bluetooth Overview

Bluetooth is a short-range, cable-replacement technology for Personal Area Networks (PANs). Operates in the ISM band (2.4–2.484 GHz). Uses frequency hopping spread spectrum at **1600 hops/second**. Data rate: **1 Mbps**. Provides high security and supports ad hoc networking via piconets.

| Radio Class | Range       | Typical Use                                  |
|-------------|-------------|----------------------------------------------|
| Class 1     | Up to 100 m | Industrial use cases                         |
| Class 2     | Up to 10 m  | Most mobile devices and consumer electronics |
| Class 3     | Up to 1–3 m | Very close-range applications                |

### Connection Establishment (3 Phases)

|                            |                                                                   |
|----------------------------|-------------------------------------------------------------------|
| <b>Inquiry / Discovery</b> | Device scans for other Bluetooth devices in its vicinity          |
| <b>Paging</b>              | Connection formed between two willing devices                     |
| <b>Connection</b>          | Device becomes active in network OR enters a low-power sleep mode |

### Operating Modes

| Mode   | Description                                                               |
|--------|---------------------------------------------------------------------------|
| Active | Fully functional — transmitting and receiving at full capacity            |
| Sniff  | Sleeps; wakes to listen at predefined intervals (power saving)            |
| Hold   | Sleeps for a defined period; then returns to active mode (power saving)   |
| Park   | Slave becomes completely inactive until master wakes it up (lowest power) |

### Bluetooth Protocol Stack

| Layer | Protocol | Function |
|-------|----------|----------|
|-------|----------|----------|

|             |                       |                                                                                        |
|-------------|-----------------------|----------------------------------------------------------------------------------------|
| Physical    | Radio                 | Frequency hopping, ISM band                                                            |
| Data Link   | Baseband              | Error correction, hop selection, security                                              |
| Data Link   | L2CAP                 | Multiplexing, segmentation/reassembly, connection-oriented and connectionless services |
| Middleware  | RFComm                | Cable replacement — emulates RS-232 serial port; up to 60 simultaneous connections     |
| Middleware  | SDP                   | Service Discovery Protocol — discovers available services and their features           |
| Application | Application protocols | Audio, file transfer, telephony, HID, etc.                                             |

### Piconets and Scatternets

- **Piconet:** Fundamental Bluetooth network unit. **1 master + up to 7 slaves**. Master controls timing via TDMA time slots and polls slaves for transmission.
- **Scatternet:** Multiple piconets working together. A device that is a slave in one piconet can act as master in another — forming a chain.
- Piconets self-configure dynamically as devices enter and leave. Use 42-bit addressing. Simplest configuration: 1 master + 1 slave (typical file/image transfer between smartphones).

## Lecture 13 · Connectivity Technologies — Part V: Z-Wave & ISA100.11a

### Z-Wave

Z-Wave is a protocol from the **Z-Wave Alliance**, designed specifically for **home automation**. Uses RF signalling with GFSK (Gaussian Frequency Shift Keying) modulation and Manchester encoding. Up to **232 nodes** per network. Each home has a unique **Home ID** — devices in one home cannot accidentally communicate with another home.

| Region | Frequency  |
|--------|------------|
| USA    | 908.42 MHz |
| Europe | 868.42 MHz |
| India  | 865.2 MHz  |

### Z-Wave Architecture

- Single **Z-Wave controller** per home manages all devices.
- Uses **mesh topology + source routing** — entire route embedded in packet before sending.
- **Direct path:** devices in direct communication range connect without relay.
- **Healing path:** multi-hop via relay nodes for out-of-range devices (bypasses walls).

### Z-Wave vs. ZigBee

| Feature  | Z-Wave                           | ZigBee                      |
|----------|----------------------------------|-----------------------------|
| Target   | Consumers (user-friendly)        | Tech experts (customisable) |
| Cost     | More expensive                   | Less expensive              |
| Battery  | Up to 7 years on one set         | Less                        |
| Adoption | 9/10 top US smart home companies | 100+ member ZigBee Alliance |

## ISA100.11a

ISA100.11a is developed by the **International Society of Automation** for industrial plants and complexes. Over **1 billion devices** use this standard. Network/Transport: TCP/UDP/IPv6. Data Link: mesh networking with frequency hopping. Radio: based on 802.15.4.

### ISA100.11a Usage Classes

| Class | Application Category                                           |
|-------|----------------------------------------------------------------|
| 0     | Emergency action (safety-critical)                             |
| 1     | Closed-loop regulatory control                                 |
| 2     | Closed-loop supervisory control                                |
| 3     | Open-loop control                                              |
| 4     | Alerting and monitoring                                        |
| 5     | Logging and downloading (no immediate operational consequence) |

## Lecture 14–15 · Sensor Networks — Parts I & II

### What is a Wireless Sensor Network (WSN)?

A WSN is a network of individual **sensor nodes** communicating via multi-hop wireless communication to enable **continuous, real-time, remote monitoring** over large terrains without human presence. Data flows hop-by-hop to a **sink node** (gateway), which forwards it to the internet.

### Sensor Node Components

| Component          | Function                                                      |
|--------------------|---------------------------------------------------------------|
| Sensing Unit       | Sensor + ADC — measures physical phenomenon                   |
| Processing Unit    | Microcontroller/processor (e.g., ATmega, ARM) — runs TinyOS   |
| Communication Unit | Transceiver radio (ZigBee, Bluetooth LE, LoRa)                |
| Power Unit         | Battery + optional energy harvesting (solar, vibration, wind) |
| Optional: GPS      | Location finding for position-aware sensing                   |

### WSN Multi-Hop Communication

Sensor nodes have short transmission ranges (ZigBee: ~30 m). Data must be relayed **hop-by-hop** through intermediate nodes acting as relays until it reaches the sink node. This is fundamental to WSN operation.

### WSN Routing Protocols

|                           |                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------|
| <b>Flooding</b>           | Broadcast received data to all neighbours — simple but energy-wasteful.                      |
| <b>Gossiping</b>          | Send to one random neighbour only — reduces overhead vs. flooding.                           |
| <b>SPIN</b>               | Data-centric — nodes negotiate what data to share before transmitting.                       |
| <b>Directed Diffusion</b> | Interest-based — sink sends queries; sources respond through reinforced paths.               |
| <b>LEACH</b>              | Hierarchical clustering — randomly rotates Cluster Head (CH) role to distribute energy load. |

### WSN Security Attacks



| Attack                  | Description                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------|
| DoS (Denial of Service) | Floods network with fake requests to exhaust resources                                          |
| Sybil attack            | Malicious node claims multiple false identities                                                 |
| Wormhole attack         | Two distant malicious nodes create a tunnel, routing data through it secretly                   |
| Selective forwarding    | Malicious node selectively drops certain packets                                                |
| Hello flood attack      | High-power attacker broadcasts hello messages; many nodes falsely believe it is their neighbour |

## WEEK 4 — Sensor Networks (III-V), UAV Networks & M2M Communication

### Lecture 16–18 · Sensor Networks — Parts III, IV & V

#### Advanced WSN Concepts

##### Data Aggregation

Combining data from multiple sensor nodes before forwarding to the sink saves bandwidth and energy. Common functions: Average, Max, Min, Sum, Count. **In-network aggregation** is performed at intermediate nodes, not at the sink.

##### Specialised Sensor Networks

| Network Type             | Standard / Technology     | Key Features                                                                    |
|--------------------------|---------------------------|---------------------------------------------------------------------------------|
| Body Area Network (WBAN) | IEEE 802.15.6             | On/in-body sensors; very low power; biocompatible; ECG, blood pressure, glucose |
| Underwater WSN (UWSN)    | Acoustic communication    | High propagation delay; high BER; ocean monitoring; tsunami detection           |
| Multimedia WSN (MWSN)    | Camera + microphone nodes | High data rates; video surveillance; traffic monitoring                         |

### Lecture 19 · UAV Networks

#### UAVs (Drones) in IoT

UAVs serve as mobile sensor nodes OR relay nodes. **FANET (Flying Ad hoc Network)** — UAVs communicate peer-to-peer. Rapid deployment, coverage of inaccessible areas, dynamic repositioning. Challenges: high mobility, frequent topology changes, energy constraints, collision avoidance.

##### UAV Applications

- Disaster response and search-and-rescue.
- Precision agriculture (crop health monitoring).
- Border surveillance and security.
- Package and medicine delivery.
- Smart city infrastructure monitoring.

### Lecture 20 · Machine to Machine (M2M) Communication

#### M2M Communication

M2M is direct communication between machines/devices **without human intervention**. M2M is a subset of IoT — IoT has a broader scope including thing-to-person and thing-to-application interactions.

| Feature      | M2M                      | IoT                                                 |
|--------------|--------------------------|-----------------------------------------------------|
| Scope        | Machine-to-machine only  | Things, people, applications — all interactions     |
| Architecture | Point-to-point typically | Large-scale network with cloud, analytics, services |

|           |                             |                                             |
|-----------|-----------------------------|---------------------------------------------|
| Standards | ETSI M2M / oneM2M, 3GPP MTC | Diverse — ZigBee, MQTT, CoAP, LoRaWAN, etc. |
|-----------|-----------------------------|---------------------------------------------|

### M2M Applications

- Smart metering (electricity, water, gas).
- Connected vehicles (real-time traffic, OBD).
- Remote patient monitoring and telemedicine.
- Industrial automation and PLCs.
- Vending machines and point-of-sale systems.

# WEEK 5 — Interoperability, Arduino (I & II) & Sensors/Actuators with Arduino

## Lecture 21 · Interoperability in IoT

### The Interoperability Challenge

IoT devices come from different vendors following different specifications, protocols, and standards. **Interoperability** is the ability of products/systems to work with other products/systems without restrictions. Without it, the vision of 'anything, anytime, anywhere' connectivity is impossible.

### Dimensions of Heterogeneity

| Dimension                     | Examples                                                              |
|-------------------------------|-----------------------------------------------------------------------|
| Communication protocols       | ZigBee (802.15.4), Bluetooth (802.15.1), WiFi (802.11), 6LoWPAN, GPRS |
| Programming languages         | JavaScript, Java, C, C++, Python, PHP                                 |
| Operating systems (IoT nodes) | TinyOS, SOS, Mantis, RETOS, vendor-specific                           |
| User-side OS                  | Windows, macOS, Linux, Ubuntu, Android, iOS                           |
| Databases                     | MySQL, Oracle, SQLite, PostgreSQL, MongoDB, InfluxDB                  |
| Data representation           | Different formats, syntaxes, semantics, languages                     |

### Types of Interoperability Needed

|                        |                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>User ↔ Device</b>   | Language, syntax, semantic barriers between a user and a device (e.g., US user accessing CCTV in Delhi and Tokyo) |
| <b>Device ↔ Device</b> | Protocol translation and semantic mapping between two heterogeneous devices                                       |
| <b>M2M / D2D / D2M</b> | Machine-to-machine, device-to-device (LTE-A D2D), device-to-machine interactions all require seamless integration |

## Lecture 22–23 · Introduction to Arduino — Parts I & II

### What is Arduino?

Arduino is an **open-source electronics platform** combining a microcontroller board with a simplified C/C++ based IDE. It is used for building interactive electronics and IoT prototypes. The popular **Arduino Uno** uses the ATmega328P microcontroller.

| Specification     | Arduino Uno Value |
|-------------------|-------------------|
| Microcontroller   | ATmega328P        |
| Operating voltage | 5V                |
| Digital I/O pins  | 14 (6 with PWM)   |
| Analog input pins | 6                 |
| Flash memory      | 32 KB             |

|             |        |
|-------------|--------|
| SRAM        | 2 KB   |
| EEPROM      | 1 KB   |
| Clock speed | 16 MHz |

### Core Arduino Functions

| Function                    | Description                                     |
|-----------------------------|-------------------------------------------------|
| pinMode(pin, MODE)          | Set pin as INPUT or OUTPUT                      |
| digitalWrite(pin, HIGH/LOW) | Write digital value to pin                      |
| digitalRead(pin)            | Read digital value from pin                     |
| analogWrite(pin, 0–255)     | Write PWM value (0=always LOW, 255=always HIGH) |
| analogRead(pin)             | Read analog value 0–1023 (10-bit ADC)           |
| delay(ms)                   | Pause execution for specified milliseconds      |
| Serial.begin(baud)          | Initialise serial communication                 |
| Serial.println(data)        | Print data to serial monitor                    |

### Communication Interfaces

|                      |                                                                            |
|----------------------|----------------------------------------------------------------------------|
| <b>UART (Serial)</b> | Pins 0 (RX) and 1 (TX); serial communication with PC or other devices      |
| <b>SPI</b>           | High-speed synchronous communication with peripherals (SD cards, displays) |
| <b>I2C</b>           | Two-wire protocol (SDA, SCL); connect multiple sensors on same bus         |

## Lecture 24–25 · Sensors & Actuators with Arduino — Parts I & II

### Common Sensor Connections

| Sensor             | Connection            | Notes                                              |
|--------------------|-----------------------|----------------------------------------------------|
| LM35 temperature   | Analog pin            | Voltage = 10mV/°C; reading × (5.0/1023 × 100) = °C |
| DHT11/DHT22        | Digital pin + library | Gives temperature AND humidity; use DHT.h library  |
| HC-SR04 ultrasonic | TRIG + ECHO pins      | Distance = (pulse_duration × 0.034) / 2 cm         |
| PIR motion sensor  | Digital pin           | Output HIGH when motion detected                   |

### Common Actuator Connections

| Actuator     | Interface                  | Notes                                                 |
|--------------|----------------------------|-------------------------------------------------------|
| LED          | Digital pin + resistor     | Current-limiting resistor required (~220Ω)            |
| Buzzer       | Digital pin                | tone(pin, frequency) function for sound               |
| Relay module | Digital pin                | Switches high-voltage AC loads from 5V Arduino signal |
| Servo motor  | Digital PWM pin            | Use Servo.h library; servo.write(angle) for 0–180°    |
| DC motor     | Motor driver (L298N/L293D) | Direction + speed (PWM) control                       |

## Wireless Communication Modules for Arduino

| Module          | Protocol            | Range              |
|-----------------|---------------------|--------------------|
| ESP8266 / ESP32 | WiFi (802.11)       | Up to 100 m indoor |
| HC-05 / HC-06   | Bluetooth Classic   | ~10 m              |
| SIM800 / SIM900 | GSM/GPRS cellular   | Nationwide         |
| LoRa (SX1276)   | LoRaWAN             | 2–15 km open area  |
| nRF24L01        | 2.4 GHz proprietary | ~100 m             |

# WEEK 6 — Python Programming, Raspberry Pi & IoT Implementation

## Lecture 26–27 · Introduction to Python Programming — Parts I & II

### Why Python for IoT?

Python is simple, readable, interpreted, and has extensive libraries for networking, data processing, hardware interfacing, and ML. It is the default language for Raspberry Pi IoT development.

### Key Python IoT Libraries

| Library      | Purpose                                 |
|--------------|-----------------------------------------|
| RPi.GPIO     | GPIO pin control on Raspberry Pi        |
| time         | Delays, timing, timestamps              |
| socket       | TCP/UDP network communication           |
| requests     | HTTP GET/POST to cloud APIs             |
| json         | Parse and generate JSON data            |
| paho-mqtt    | MQTT publish/subscribe client           |
| Adafruit_DHT | DHT11/DHT22 sensor interface            |
| picamera     | Raspberry Pi camera control             |
| flask        | Lightweight web server / REST API on Pi |

### Python IoT Code Patterns

#### Reading a sensor and publishing via MQTT:

```
import paho.mqtt.client as mqtt, json, time, Adafruit_DHT
client = mqtt.Client()
client.connect('broker_address', 1883)
while True:
    hum, temp = Adafruit_DHT.read_retry(Adafruit_DHT.DHT11, 4)
    client.publish('sensor/data', json.dumps({'temp': temp, 'hum': hum}))
    time.sleep(5)
```

## Lecture 28–29 · Introduction to Raspberry Pi — Parts I & II

### Raspberry Pi vs. Arduino

| Feature | Raspberry Pi 3B                  | Arduino Uno           |
|---------|----------------------------------|-----------------------|
| Type    | Single-board computer            | Microcontroller board |
| OS      | Linux (Raspberry Pi OS)          | None — bare metal     |
| CPU     | 1.2 GHz quad-core ARM Cortex-A53 | 16 MHz ATmega328P     |
| RAM     | 1 GB                             | 2 KB SRAM             |
| Storage | MicroSD card                     | 32 KB Flash           |

|              |                                     |                                  |
|--------------|-------------------------------------|----------------------------------|
| Languages    | Python, C, Java, Node.js, etc.      | C / C++ (Arduino IDE)            |
| GPIO         | 40 pins (3.3V logic)                | 14 digital + 6 analog (5V logic) |
| Connectivity | WiFi, Bluetooth, Ethernet           | None built-in (use shields)      |
| Power draw   | ~700 mA                             | ~50 mA                           |
| Best for     | Complex IoT gateway, edge computing | Simple sensor/actuator control   |

### Important GPIO Note

Raspberry Pi GPIO pins operate at **3.3V** (not 5V like Arduino). Connecting 5V signals directly to GPIO pins can permanently damage the Pi. Always use level shifters when interfacing with 5V sensors/actuators.

## Lecture 30 · Implementation of IoT with Raspberry Pi — Part I

### Raspberry Pi as IoT Gateway

The Raspberry Pi can collect data from multiple sensor nodes (via ZigBee coordinator, serial, WiFi, BLE), aggregate data, perform local edge analytics, and forward results to the cloud.

### Example: Flask REST API for IoT Dashboard

```
from flask import Flask, jsonify
app = Flask(__name__)
@app.route('/data')
def get_data():
    return jsonify({'temperature': 25.5, 'humidity': 60})
app.run(host='0.0.0.0', port=5000)
```

Access the dashboard from any browser on the local network: **http://<pi-ip>:5000/data**



# WEEK 7 — Raspberry Pi IoT (II-III), SDN (I-II) & SDN-IoT Networking

## Lecture 31–32 · Implementation of IoT with Raspberry Pi — Parts II & III

### Advanced Raspberry Pi IoT

#### Database Integration

|          |                                                                                 |
|----------|---------------------------------------------------------------------------------|
| SQLite   | Lightweight local database — ideal for storing sensor readings on the Pi        |
| InfluxDB | Purpose-built time-series database — fast ingestion and querying of sensor data |
| Grafana  | Real-time visualisation dashboard — connects to InfluxDB; beautiful charts      |

#### OpenCV for Smart Vision Applications

- Image processing for smart security cameras on Raspberry Pi.
- Face detection, motion detection using camera module.
- Alert sent via MQTT or email on intrusion detection.

## Lecture 33–34 · Software Defined Networking — Parts I & II

### What is SDN?

Software Defined Networking **decouples the control plane** (routing decisions) from the **data plane** (packet forwarding). The control plane is moved to a **centralised SDN Controller**. Network behaviour becomes programmable via software.

#### SDN Architecture (3 Layers)

| Layer                          | Description                                                                |
|--------------------------------|----------------------------------------------------------------------------|
| Application Layer              | Network applications — routing, load balancing, QoS, security policies     |
| Control Layer (SDN Controller) | Logical centralised brain; manages flow tables; installs rules in switches |
| Infrastructure Layer           | Physical switches and routers that execute flow rules (data plane only)    |

#### OpenFlow Protocol

**OpenFlow** is the communication protocol between SDN controller and switches. The controller installs **flow table entries** in switches. Each entry: *Match fields* (IP src/dst, port) → *Actions* (forward, drop, modify header).

- **Proactive:** controller pre-installs rules before traffic arrives.
- **Reactive:** switch sends Packet\_In to controller for first packet of each new flow.

#### SDN Benefits for IoT

- ◆ Centralised management of all IoT devices from one controller.
- ◆ Dynamic network reconfiguration without touching hardware.
- ◆ Vendor-agnostic hardware — commodity switches work.
- ◆ Enables network virtualisation and multi-tenancy.
- ◆ Significantly reduces CAPEX and OPEX.

## Lecture 35 · Software-Defined IoT Networking — Part I

## SDN for Wireless Sensor Networks (WSN)

Traditional WSN limitations: manual per-device configuration, static routing, difficult in-field reprogramming, protocol-specific solutions. SDN brings centralised control, dynamic reprogramming, and simplified management of large-scale WSN deployments.

### SDN-WSN Approaches (Chronological)

| Approach        | Year  | Description                                                                        |
|-----------------|-------|------------------------------------------------------------------------------------|
| Sensor OpenFlow | 2012  | First OpenFlow-based WSN control (Luo et al., IEEE Communications Letters)         |
| Soft-WSN        | ~2014 | Proposed by Prof. Misra's SWAN Group at IIT KGP; tested on real sensor hardware    |
| SDN-WISE        | Later | Richer flow table support for resource-constrained nodes; current state-of-the-art |

# WEEK 8 — SDN-IoT (II) & Cloud Computing — Fundamentals to Case Studies

## Lecture 36 · Software-Defined IoT Networking — Part II

### SDN for Mobile IoT Networks

Traditional mobile network problems: static over-provisioning, inflexibility, high cost (CAPEX + OPEX), difficulty managing heterogeneous devices.

### Key SDN Solutions for Mobile IoT

| Solution  | Published     | Key Contribution                                                                    |
|-----------|---------------|-------------------------------------------------------------------------------------|
| SDWMN     | ONF           | Virtualised Radio Resource Management Unit for interference control between eNodeBs |
| ODIN      | Literature    | Agent-based — ODIN agent at access point + ODIN master at SDN controller            |
| Ubi-Flow  | INFOCOM 2015  | Mobility management + flow scheduling with network partitioning and load balancing  |
| Mobi-Flow | Globecom 2016 | Mobility-aware flow rule placement for mobile IoT (proposed by SWAN Group, IIT KGP) |

## Lecture 37 · Cloud Computing — Fundamentals

### NIST Definition of Cloud Computing

Cloud computing provides on-demand access to a shared pool of configurable computing resources (networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort.

### Five Essential Characteristics

- **On-demand self-service:** Resources provisioned automatically without human provider intervention.
- **Broad network access:** Available over any network from any standard device.
- **Resource pooling:** Multi-tenant model — resources shared among multiple customers.
- **Rapid elasticity:** Resources scale up/down quickly to meet demand.
- **Measured service:** Usage metered and billed on pay-per-use basis.

### Cloud Deployment Models

| Model     | Description                                              | IoT Use                                           |
|-----------|----------------------------------------------------------|---------------------------------------------------|
| Public    | Shared multi-tenant; owned by provider (AWS, Azure, GCP) | IoT backend for startups and SMEs                 |
| Private   | Dedicated to one organisation; on-premises or hosted     | Enterprise IoT with strict data privacy           |
| Hybrid    | Combination of public + private clouds                   | Sensitive data on-prem; analytics on public cloud |
| Community | Shared by organisations with common concerns             | Healthcare IoT consortium, smart city consortium  |

## Lecture 38 · Cloud Computing — Service Models

### Three Cloud Service Models

| Model                              | Provider manages                     | User manages                        | IoT Examples                                              |
|------------------------------------|--------------------------------------|-------------------------------------|-----------------------------------------------------------|
| IaaS (Infrastructure as a Service) | Hardware, virtualisation, networking | OS, runtime, middleware, apps, data | AWS EC2, Google Compute Engine — host IoT backends        |
| PaaS (Platform as a Service)       | Hardware + OS + middleware           | Applications and data only          | Google App Engine, Heroku — rapid IoT app development     |
| SaaS (Software as a Service)       | Everything                           | Just use the application            | ThingSpeak, AWS IoT Core dashboard — ready-made IoT tools |

### Major IoT Cloud Platforms

| Platform              | Provider  | Key Feature                                                  |
|-----------------------|-----------|--------------------------------------------------------------|
| AWS IoT Core          | Amazon    | Managed MQTT broker; device shadows; Lambda integration      |
| Azure IoT Hub         | Microsoft | Device twin concept; MQTT/AMQP/HTTPS; Azure Stream Analytics |
| Google Cloud IoT Core | Google    | Pub/Sub based; BigQuery + ML integration                     |
| IBM Watson IoT        | IBM       | Enterprise IoT; AI-driven analytics                          |
| ThingSpeak            | MathWorks | Open platform; MATLAB visualisation; free tier available     |

## Lecture 39–40 · Cloud Computing — Security, Management & Case Studies

### Cloud Security for IoT

| Threat               | Countermeasure                                                          |
|----------------------|-------------------------------------------------------------------------|
| Data privacy         | Encryption at rest (AES) and in transit (TLS/SSL)                       |
| Unauthorised access  | Identity and Access Management (IAM); Multi-Factor Authentication (MFA) |
| DDoS attacks         | Cloud CDN, rate limiting, auto-scaling                                  |
| Device impersonation | X.509 certificates for each IoT device                                  |
| Insider threats      | Audit logging, access monitoring, least-privilege policies              |

### Cloud IoT Case Studies

- **Smart Metering:** Smart meters send consumption every 15 min → cloud analytics → billing, outage detection, customer portal.
- **Connected Healthcare:** Wearables stream ECG/SpO2 to cloud → ML anomaly detection → doctor alerts in real time.
- **Precision Agriculture:** Soil sensors + weather stations → cloud ML → irrigation/fertilisation recommendations to farmers.
- **Predictive Maintenance:** Factory machines report vibration/temperature → cloud analytics predicts failures → proactive maintenance scheduling.

# WEEK 9 — Cloud Practical, Sensor-Cloud & Fog Computing

## Lecture 41 · Cloud Computing — Practical

### ThingSpeak: IoT Data Platform

ThingSpeak is a free IoT cloud platform for sensor data visualisation and analytics. Channels group related sensor fields; data uploaded via REST API; MATLAB-based analysis built in.

```
import requests
API_KEY = 'YOUR_THINGSPEAK_KEY'
temp, hum = 25.5, 60
url = f'https://api.thingspeak.com/update?api_key={API_KEY}&field1={temp}&field2={hum}'
r = requests.get(url)
print('HTTP status:', r.status_code)
```

## Lecture 42–43 · Sensor-Cloud — Parts I & II

### Sensor-Cloud Infrastructure

Sensor-Cloud integrates physical wireless sensor networks with cloud computing. Sensors collect data → transmitted to cloud → cloud provides storage, processing, and analytics as services. Users access sensor data through the cloud without owning the physical sensors.

#### Three-Layer Architecture

|               |                                                      |
|---------------|------------------------------------------------------|
| Sensor Layer  | Physical sensor nodes deployed in the field          |
| Gateway Layer | Aggregates data from sensor nodes; forwards to cloud |
| Cloud Layer   | Storage, processing, analytics, user interface       |

#### Virtual Sensors — Key Concept

A **virtual sensor** is an abstraction of one or more physical sensors. Users subscribe to virtual sensors rather than physical ones. Multiple virtual sensors can be created from one physical sensor; one physical sensor can contribute to multiple virtual sensors. This decouples users from physical sensor management.

#### Sensor-as-a-Service (Senaas)

Cloud business model where sensor data is offered as a paid service via APIs. Sensor owners expose data; applications subscribe and pay for access. Similar to SaaS but for physical-world sensing.

## Lecture 44–45 · Fog Computing — Parts I & II

### What is Fog / Edge Computing?

Fog computing (Cisco term) / Edge computing extends cloud capabilities to the **edge of the network**, closer to IoT devices. Instead of sending all data to distant cloud, processing is done at local fog nodes. Essential for latency-sensitive and bandwidth-constrained applications.

#### Cloud vs. Fog Comparison

| Feature  | Cloud                             | Fog/Edge                   |
|----------|-----------------------------------|----------------------------|
| Location | Centralised (remote data centres) | Distributed (network edge) |

|                 |                                    |                                               |
|-----------------|------------------------------------|-----------------------------------------------|
| Latency         | High (100s of ms)                  | Low (1–10 ms)                                 |
| Bandwidth usage | High — all raw data sent upstream  | Low — only processed results sent to cloud    |
| Scale           | Virtually unlimited (auto-scaling) | Limited per fog node                          |
| Best for        | Long-term storage, deep analytics  | Real-time control, time-critical applications |
| Privacy         | Data leaves premises               | Sensitive data processed locally              |

### Computing Continuum

**Cloud** (remote data centres) → **Fog** (edge routers, gateways, local servers) → **Mist** (computing on the sensor/device itself). Each layer handles different latency and bandwidth requirements.

### Major Fog Platforms

| Platform       | Provider  | Description                                                    |
|----------------|-----------|----------------------------------------------------------------|
| AWS Greengrass | Amazon    | Cloud-managed edge computing; runs Lambda functions locally    |
| Azure IoT Edge | Microsoft | Deploy cloud workloads (ML models, containers) to edge devices |
| OpenFog        | IEEE 1934 | Open standard for fog computing architecture                   |

### Fog Computing Applications

- **Autonomous vehicles:** Split-second collision avoidance decisions — no time for cloud round-trip.
- **Smart grid:** Local fault detection and immediate load-shedding response.
- **Healthcare:** Critical patient monitoring with instant alerts — cannot wait for cloud.
- **Smart manufacturing:** Real-time quality control defect detection on the production line.

# WEEK 10 — Smart Cities & Smart Homes, Connected Vehicles

## Lecture 46–48 · Smart Cities and Smart Homes — Parts I, II & III

### Smart City Vision

A smart city uses **ICT and IoT** to improve services, resource efficiency, and quality of life. IoT is a foundational enabling technology, interconnecting transportation, energy, water, healthcare, buildings, and governance.

### Smart City Components

| Domain         | IoT Application                                                             |
|----------------|-----------------------------------------------------------------------------|
| Transportation | Adaptive traffic lights, smart parking (fill sensors), real-time navigation |
| Energy         | Smart grid, smart street lighting (presence-based), renewable integration   |
| Water          | Leak detection, quality monitoring, consumption metering                    |
| Waste          | Fill-level sensors in bins → optimal collection routing                     |
| Healthcare     | Emergency response optimisation, air quality monitoring                     |
| Public Safety  | CCTV with ML analytics, emergency vehicle coordination                      |
| Environment    | Air quality, noise mapping, flood/landslide monitoring                      |

### Smart Home Architecture

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| <b>Sensors + Actuators</b> | Smart lights, thermostat, door locks, cameras, appliances           |
| <b>Local Hub / Gateway</b> | ZigBee/Z-Wave coordinator (Amazon Echo, Google Nest, Apple HomePod) |
| <b>Cloud Backend</b>       | Remote access, automation rules, ML-based recommendations           |
| <b>User Application</b>    | Smartphone app or voice assistant (Alexa, Google, Siri)             |

### Smart Home Standards Ecosystem

| Standard/Platform | Backer                      | Notes                                                  |
|-------------------|-----------------------------|--------------------------------------------------------|
| Amazon Alexa      | Amazon                      | Voice control; 100,000+ compatible devices             |
| Google Home       | Google                      | Google Assistant; Nest product line                    |
| Apple HomeKit     | Apple                       | iOS-based; strong security focus; Secure Enclave       |
| Matter            | Apple/Google/Amazon/Samsung | New unified cross-platform smart home standard (2022+) |

## Lecture 49–50 · Connected Vehicles — Parts I & II

### Vehicle-to-Everything (V2X) Communication

| Type | Full Name | Description |
|------|-----------|-------------|
|------|-----------|-------------|

|     |                           |                                                               |
|-----|---------------------------|---------------------------------------------------------------|
| V2V | Vehicle-to-Vehicle        | Safety messages between nearby vehicles (collision avoidance) |
| V2I | Vehicle-to-Infrastructure | Communicate with traffic lights, road signs, toll gates       |
| V2N | Vehicle-to-Network        | Cellular connectivity for maps, infotainment, OTA updates     |
| V2P | Vehicle-to-Pedestrian     | Warn pedestrians of approaching vehicles                      |

### Connected Vehicle Technologies

|                              |                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------|
| <b>DSRC / 802.11p (WAVE)</b> | Dedicated Short-Range Communications; dedicated spectrum; very low latency for safety applications |
| <b>C-V2X (LTE)</b>           | Cellular Vehicle-to-Everything; leverages existing LTE network; longer range                       |
| <b>5G NR-V2X</b>             | Next generation; ultra-low latency for fully autonomous driving coordination                       |

### SAE Levels of Autonomy

| Level | Name                   | Description                                               |
|-------|------------------------|-----------------------------------------------------------|
| L0    | No automation          | Human controls everything                                 |
| L1    | Driver assistance      | Adaptive cruise control or lane keeping (one at a time)   |
| L2    | Partial automation     | System handles both steering and speed (Tesla Autopilot)  |
| L3    | Conditional automation | System drives; human must be ready to take over           |
| L4    | High automation        | No driver needed in defined geographic areas / conditions |
| L5    | Full automation        | No driver needed anywhere, in any conditions              |

### Sensors in Autonomous Vehicles

- **LiDAR:** 360° 3D point cloud of surroundings; most accurate for obstacle detection.
- **RADAR:** Works in all weather (rain, fog); measures speed of objects.
- **Cameras:** Lane detection, traffic sign and traffic light recognition.
- **GPS/GNSS:** Global positioning (accurate to ~1 m with DGPS).
- **IMU (Inertial Measurement Unit):** Acceleration and orientation (dead reckoning between GPS fixes).



# WEEK 11 — Smart Grid, Industrial IoT & Data Handling and Analytics (Part I)

## Lecture 51–52 · Smart Grid — Parts I & II

### Traditional Grid vs. Smart Grid

| Feature       | Traditional Grid                   | Smart Grid                                             |
|---------------|------------------------------------|--------------------------------------------------------|
| Communication | Unidirectional (source → consumer) | Bidirectional (source ↔ consumer)                      |
| Monitoring    | Manual, infrequent                 | Real-time, automated, continuous                       |
| Restoration   | Manual intervention required       | Self-healing — automatic fault isolation               |
| Consumer role | Passive consumer only              | Active participant — demand response, micro-generation |
| Pricing       | Fixed flat rate                    | Dynamic time-of-use pricing                            |
| Renewables    | Difficult to integrate             | Seamless integration of solar, wind, EVs               |
| Energy theft  | Difficult to detect                | Tamper detection via smart metering                    |

The smart grid is conceptualised as a **Cyber-Physical System (CPS)** — the cyber (network/ICT) component is tightly integrated with the physical (power) component. Also called: *Electricity with a Brain*, *Energy Internet*, *Electro-Net*.

### Smart Grid Architecture Components

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| <b>Generation</b>   | Renewable (solar, wind) + non-renewable (coal, nuclear) power plants      |
| <b>Transmission</b> | High-voltage lines carrying bulk power                                    |
| <b>Substations</b>  | Transform and distribute to neighbourhoods                                |
| <b>AMI</b>          | Advanced Metering Infrastructure — network of programmable smart meters   |
| <b>HAN</b>          | Home Area Network — ZigBee/Z-Wave connecting smart appliances             |
| <b>NAN</b>          | Neighbourhood Area Network — connects smart meters to data aggregator     |
| <b>MDMS</b>         | Meter Data Management System — cloud-based meter data analytics           |
| <b>SCADA/DMS</b>    | Supervisory Control and Data Acquisition / Distribution Management System |

#### Smart Grid Benefits

- ◆ Customers: real-time usage monitoring, programmable smart appliances, EV charging integration, time-of-use pricing options.
- ◆ Utilities: reduced blackouts/brownouts, faster fault restoration, integration of micro-grids and renewables.
- ◆ Society: reduced electricity theft, improved grid resilience, support for distributed generation.

## Lecture 53–54 · Industrial Internet of Things — Parts I & II

### IIoT and Industry 4.0

IIoT applies IoT in industrial settings (factories, plants, utilities, logistics). **Industry 4.0** — the fourth industrial revolution — combines CPS, IoT, Big Data, AI, and additive manufacturing. IIoT bridges **OT (Operational Technology)** — legacy SCADA, PLCs, DCS — with **IT (Information Technology)** — cloud, analytics, ERP.

### IIoT vs. Consumer IoT

| Feature     | Consumer IoT           | IIoT                                         |
|-------------|------------------------|----------------------------------------------|
| Environment | Home / office          | Factory / industrial plant                   |
| Reliability | Best-effort acceptable | Mission-critical — near 100% uptime required |
| Latency     | Can tolerate seconds   | Often < 1 ms required                        |
| Protocols   | WiFi, BLE, MQTT        | HART, ISA100.11a, Modbus, OPC-UA, PROFINET   |
| Device life | 2–5 years              | 10–30 years                                  |
| Safety      | Important              | Safety-critical — IEC 61508 / IEC 62443      |

### Industry 4.0 Pillars

| Technology                    | Description                                                                        |
|-------------------------------|------------------------------------------------------------------------------------|
| Cyber-Physical Systems (CPS)  | Machines with embedded sensors/actuators connected to internet                     |
| Digital Twin                  | Real-time virtual replica of a physical asset; simulation and predictive analytics |
| Big Data Analytics            | Deriving production insights from large volumes of machine data                    |
| Artificial Intelligence       | Autonomous decision-making in production processes                                 |
| Collaborative Robots (Cobots) | Robots working safely alongside human workers                                      |
| Additive Manufacturing        | 3D printing for on-demand custom spare parts                                       |
| Augmented / Virtual Reality   | Remote maintenance guidance, operator training                                     |

### Key Industrial Protocols

| Protocol          | Use                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------|
| Modbus RTU/TCP    | Simple serial/Ethernet protocol for PLCs; widely used legacy standard                   |
| PROFIBUS/PROFINET | German fieldbus standard for factory automation                                         |
| OPC-UA            | Platform-independent, secure, modern IIoT standard; supports semantic information model |
| MQTT / AMQP       | Increasingly adopted for IIoT cloud integration                                         |

## Lecture 55 · Data Handling and Analytics — Part I

### Big Data in IoT

IoT generates massive continuous data streams. Characterised by the **5 V's**:

| V        | Meaning             | IoT Context                                  |
|----------|---------------------|----------------------------------------------|
| Volume   | Massive quantity    | Trillions of sensor readings daily           |
| Velocity | Speed of generation | Real-time streaming from millions of devices |

|          |                            |                                                                                 |
|----------|----------------------------|---------------------------------------------------------------------------------|
| Variety  | Diversity of data types    | Structured (numbers), unstructured (images, audio), semi-structured (JSON, XML) |
| Veracity | Data quality and accuracy  | Sensor noise, failures, calibration drift                                       |
| Value    | Business/operational value | Extracting actionable insights from raw data                                    |

### IoT Data Pipeline

- **Collection:** Sensors → MQTT/CoAP → broker
- **Ingestion:** Stream to data platform (Apache Kafka, AWS Kinesis)
- **Storage:** Time-series DB (InfluxDB), NoSQL (Cassandra, MongoDB), Data Lake (AWS S3)
- **Processing:** Stream analytics (Apache Flink, Spark Streaming) OR batch ML (Hadoop)
- **Visualisation:** Grafana, Kibana, Power BI dashboards
- **Action:** Actuator commands, alerts, business intelligence reports

### Time-Series Databases

Most IoT data is time-stamped → **time-series data**. Optimised databases: **InfluxDB** (open-source, fast ingestion), **TimescaleDB** (PostgreSQL extension), **Amazon Timestream**. Common operations: average over time window, spike detection, downsampling.

# WEEK 12 — Data Analytics (II) & Case Studies: Agriculture, Healthcare, Activity Monitoring

## Lecture 56 · Data Handling and Analytics — Part II

### Machine Learning for IoT Analytics

| ML Type                     | Applications in IoT                                      |
|-----------------------------|----------------------------------------------------------|
| Supervised (Classification) | Anomaly detection: normal vs. faulty machine operation   |
| Supervised (Regression)     | Forecasting: predict future temperature, energy demand   |
| Unsupervised (Clustering)   | Group similar sensor reading patterns (k-means)          |
| Anomaly Detection           | Identify unusual patterns in sensor data streams         |
| Reinforcement Learning      | HVAC control optimisation; autonomous vehicle navigation |

### Deep Learning for IoT

| Model                              | IoT Application                                                            |
|------------------------------------|----------------------------------------------------------------------------|
| CNN (Convolutional Neural Network) | Image classification for visual inspection; activity recognition           |
| LSTM (Long Short-Term Memory)      | Time-series forecasting — predict equipment failure from sensor history    |
| Autoencoder                        | Anomaly detection in sensor streams — large reconstruction error = anomaly |
| CNN + LSTM hybrid                  | Activity recognition from raw accelerometer/gyroscope data                 |

### Edge/Federated Learning

|                           |                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------|
| <b>TensorFlow Lite</b>    | Compressed ML models for microcontrollers — run inference on Arduino, Raspberry Pi                     |
| <b>Edge Impulse</b>       | Platform for training and deploying ML models on embedded devices                                      |
| <b>Federated Learning</b> | Train ML models across distributed devices without centralising raw personal data — privacy-preserving |

## Lecture 57 · Case Study: Smart Agriculture

### IoT-Based Precision Agriculture

Addresses challenges of traditional farming: inefficient water use, over-fertilisation, lack of real-time crop health monitoring. IoT enables **precision agriculture** — applying exactly the right inputs at the right time and place.

### IoT Sensors in Agriculture

| Sensor / Device      | Measurement / Function                                                |
|----------------------|-----------------------------------------------------------------------|
| Soil moisture sensor | Detects when soil is too dry → triggers irrigation                    |
| NPK sensor           | Measures nitrogen, phosphorus, potassium levels → fertiliser guidance |
| Weather station      | Local temperature, humidity, rainfall, wind → microclimate monitoring |

|                   |                                                               |
|-------------------|---------------------------------------------------------------|
| Drone with camera | NDVI (crop health index), pest and disease detection from air |
| Water flow sensor | Monitors irrigation system for leaks or blockages             |
| Livestock GPS tag | Tracks cattle location and activity for herd management       |

### Architecture

- **Field sensors** → ZigBee/LoRa → Raspberry Pi/Arduino gateway
- **Gateway** → 3G/4G/WiFi → Cloud (ThingSpeak / AWS IoT)
- **Cloud analytics** → irrigation schedule, yield prediction, pest alerts
- **Farmer app** → smartphone notifications and remote actuator control

#### Smart Agriculture Impact

- ◆ Water savings: up to 30–50% compared to traditional flood irrigation.
- ◆ Fertiliser efficiency: 20–30% reduction through precision application.
- ◆ Yield improvement: up to 20% through early disease/pest detection.
- ◆ Labour cost reduction via automated monitoring and actuation.

### LoRa for Agriculture

LoRa (Long Range) is ideal for large farms. Range: 2–15 km in open areas. Very low power — batteries last years. Very low data rate — suitable for sensor readings (not video). **LoRaWAN** is the network protocol for LoRa devices, using a star topology with network servers.

## Lecture 58 · Case Study: Smart Healthcare

### IoT in Healthcare Applications

| Application               | IoT Devices / Technology                                                        | Benefit                                               |
|---------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| Remote Patient Monitoring | ECG patches, SpO2, blood pressure, glucose wearables + smartphone gateway       | Continuous monitoring at home; reduce hospitalisation |
| Smart Hospital            | RFID tracking of patients/staff/equipment; smart beds; connected infusion pumps | Operational efficiency; patient safety                |
| Telemedicine              | Video consultation + wearable sensor data integration                           | Healthcare access in remote areas                     |
| Implantable IoT           | Smart pacemakers, cochlear implants, continuous glucose monitors (CGM)          | Continuous therapeutic monitoring                     |
| Drug delivery             | IoT-enabled infusion pumps with auto-dosing and alerts                          | Prevent medication errors                             |

### Healthcare IoT Pipeline

- **Wearable sensors** → Bluetooth LE → Smartphone gateway
- **Smartphone** → 4G/WiFi → Health cloud (AWS HealthLake, Azure Health API)
- **Cloud ML** → Anomaly detection (arrhythmia, oxygen drop, glucose spike)
- **Healthcare portal** → Doctor dashboard and alert system
- **Emergency response** → Automated ambulance dispatch for critical events

### Healthcare IoT Security Requirements

- **Data privacy:** Patient data is highly sensitive (HIPAA in US, IT Act in India).

- **End-to-end encryption:** All patient data encrypted in transit and at rest.
- **Device authentication:** X.509 certificates prevent rogue device injection.
- **Access control:** Only authorised healthcare providers access patient data.

## Lecture 59–60 · Case Study: Activity Monitoring — Parts I & II

### Human Activity Recognition (HAR)

HAR detects and classifies human physical activities using wearable or ambient sensors. Applications: elderly fall detection, fitness tracking, workplace safety, physiotherapy rehabilitation, gesture-based control.

#### Sensor Modalities for Activity Recognition

| Sensor         | Measurement                | Activity Insight                       |
|----------------|----------------------------|----------------------------------------|
| Accelerometer  | 3-axis linear acceleration | Step counting, posture, fall detection |
| Gyroscope      | 3-axis rotational motion   | Rotation, orientation changes          |
| Magnetometer   | Magnetic field (compass)   | Absolute orientation, navigation       |
| Barometer      | Atmospheric pressure       | Altitude change — stair detection      |
| GPS            | Position, speed            | Outdoor route, distance, speed         |
| PPG            | Blood volume pulse (wrist) | Heart rate, HRV, stress level          |
| EMG            | Muscle electrical activity | Gesture recognition, muscle fatigue    |
| Camera / Depth | Visual frames, depth map   | Whole-body pose estimation             |

#### HAR Processing Pipeline

|                       |                                                                                |
|-----------------------|--------------------------------------------------------------------------------|
| 1. Data Collection    | Sample sensors at 50–100 Hz; collect ground-truth labels                       |
| 2. Preprocessing      | Remove noise (low-pass filter), normalise values, handle missing data          |
| 3. Segmentation       | Sliding window (e.g., 2-second windows with 50% overlap)                       |
| 4. Feature Extraction | Time domain: mean, variance, zero-crossing. Frequency domain: FFT coefficients |
| 5. Classification     | SVM, Random Forest (classic ML) OR CNN+LSTM (deep learning)                    |
| 6. Post-processing    | Majority vote over window predictions; output activity label                   |

#### Fall Detection for Elderly

Falls are the leading cause of injury-related death in elderly (65+). Detection approach: sudden acceleration spike → brief freefall → impact → prolonged inactivity → classify as fall. Alert sent to caregiver/emergency services via smartphone. ML-based approach reduces false positives compared to simple threshold methods.

#### Gait Analysis

Study of walking pattern using smartphone accelerometer or smart insoles. Detects early signs of Parkinson's disease, dementia, or rehabilitation progress. Features: step length, cadence, stride symmetry.

#### Privacy in Activity Monitoring

- **On-device processing:** Run ML model on the wearable; only activity labels (not raw sensor data) sent to cloud.
- **Federated learning:** Train HAR models across many users without centralising personal activity data.
- User **consent and transparency** are legally and ethically essential.

# WEEK — MASTER REFERENCE TABLES

## All 60 Lectures — Week-wise Index

| Week | Lectures | Topics Covered                                                                           |
|------|----------|------------------------------------------------------------------------------------------|
| 1    | 1–5      | IoT intro (I & II), Sensing, Actuation, IoT Networking basics (I)                        |
| 2    | 6–10     | MQTT, SMQTT, CoAP, XMPP, AMQP; 802.15.4, ZigBee, 6LoWPAN, RFID                           |
| 3    | 11–15    | HART/WirelessHART, NFC, Bluetooth, Z-Wave, ISA100.11a; Sensor Networks (I & II)          |
| 4    | 16–20    | Sensor Networks (III–V), UAV Networks, M2M Communication                                 |
| 5    | 21–25    | Interoperability; Arduino (I & II); Sensors+Actuators with Arduino (I & II)              |
| 6    | 26–30    | Python (I & II); Raspberry Pi (I & II); IoT with Raspberry Pi (I)                        |
| 7    | 31–35    | IoT with Raspberry Pi (II & III); SDN (I & II); SDN-IoT Networking (I)                   |
| 8    | 36–40    | SDN-IoT (II); Cloud Fundamentals, Service Models, Security & Case Studies                |
| 9    | 41–45    | Cloud Practical; Sensor-Cloud (I & II); Fog Computing (I & II)                           |
| 10   | 46–50    | Smart Cities & Homes (I–III); Connected Vehicles (I & II)                                |
| 11   | 51–55    | Smart Grid (I & II); Industrial IoT (I & II); Data Analytics (I)                         |
| 12   | 56–60    | Data Analytics (II); Case Studies: Agriculture, Healthcare, Activity Monitoring (I & II) |

## Complete IoT Protocol Reference

| Protocol      | OSI Layer(s)          | Transport | Primary Use Case                                       |
|---------------|-----------------------|-----------|--------------------------------------------------------|
| MQTT          | Application           | TCP       | Lightweight publish-subscribe IoT messaging            |
| SMQTT         | Application           | TCP       | Secure MQTT with broadcast encryption                  |
| CoAP          | Session / Application | UDP       | Constrained device web communication (HTTP equivalent) |
| XMPP          | Application           | TCP       | Real-time messaging, IoT, cloud environments           |
| AMQP          | Application           | TCP       | Enterprise/business messaging and queuing              |
| IEEE 802.15.4 | PHY + MAC             | —         | Foundation for ZigBee, 6LoWPAN, WirelessHART           |
| ZigBee        | Network – Application | —         | WSN, home automation, smart energy (AODV routing)      |
| 6LoWPAN       | Network               | —         | IPv6 over low-power wireless IoT devices               |
| RFID          | PHY                   | —         | Object/asset identification via radio tags             |
| NFC           | PHY                   | —         | Very short range (<20 cm) payment, tagging             |
| Bluetooth     | PHY – Application     | —         | Short-range cable replacement; piconet/scatternet      |
| WirelessHART  | PHY – Application     | —         | Industrial IoT; process automation                     |
| Z-Wave        | PHY – Application     | —         | Home automation (868/908 MHz)                          |

|            |                   |     |                                                     |
|------------|-------------------|-----|-----------------------------------------------------|
| ISA100.11a | PHY – Application | —   | Industrial plant automation (IEC standard)          |
| LoRaWAN    | MAC – Network     | —   | Long-range, low-power IoT (agriculture, smart city) |
| OpenFlow   | SDN Control Plane | TCP | SDN controller-to-switch communication              |

## Cloud Service Model Comparison

| Model | You manage                 | Provider manages             | Examples                                         |
|-------|----------------------------|------------------------------|--------------------------------------------------|
| IaaS  | OS, middleware, apps, data | Hardware, virtualisation     | AWS EC2, Azure VMs, Google Compute               |
| PaaS  | Apps and data only         | Hardware + OS + middleware   | Heroku, Google App Engine, AWS Elastic Beanstalk |
| SaaS  | Just use the application   | Everything                   | ThingSpeak, Salesforce, Office 365               |
| FaaS  | Function code only         | All infrastructure + runtime | AWS Lambda, Azure Functions                      |

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*End of Notes — Introduction to Internet of Things | Prof. Sudip Misra | IIT Kharagpur | NPTEL*